Preliminary On-Orbit Performance Test of the First Polarimetric Synchronization Monitoring Atmospheric Corrector (SMAC) On-Board High-Spatial Resolution Satellite Gao Fen Duo Mo (GFDM)

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***Abstract*— Obtaining accurate atmospheric parameters, e.g., aerosol optical depth (AOD) and column water vapor (CWV), is important for the quantitative atmospheric correction (AC) of the high-spatial resolution remote sensing images. However, due to the strong temporal and spatial changes of the atmospheric parameters, it will be a challenge to ensure spatiotemporal registration of the satellite images given the AC parameters obtained separately from ground-based or other satellite products, which affects significantly the accuracy of the AC. The China National Space Administration launched a high resolution and multimode imaging satellite [Gao Fen Duo Mo (GFDM)] in July 2020, which has multifunctional observation modes and flexible mobility, with a high-spatial resolution imaging sensor (0.42 m in panchromatic and 1.6 m in multispectrum) and equipped the synchronization monitoring atmospheric corrector (SMAC) sensor. As the first atmospheric corrector with polarization detection capability on-board high-spatial resolution satellite, SMAC is designed to obtain multispectral intensity and polarized data and to retrieve synchronously AC parameters in the same field of view with main sensor. Based on the SMAC in-orbit test data, a lookup table method using the optimized inversion framework and a dual-channel ratio retrieval method are developed to derive AOD and CWV, respectively, in this article. The AOD and CWV results are validated against the AERosol RObotic NETwork (AERONET). The preliminary test of AC performance on the multispectral images of GFDM satellite indicates that SMAC is of great potential to improve the quality of the main sensor's image.**

***Index Terms*— Aerosol optical depth (AOD), atmospheric correction (AC), column water vapor (CWV), Gao Fen Duo Mo (GFDM), polarization, synchronization monitoring atmospheric corrector (SMAC).**

I. Introduction

HIGH-SPATIAL resolution remote sensing images provide rich spatial details and have been widely used in precision agriculture, disaster prevention and mitigation, urban planning, environmental protection, and other fields [6], [31], [32], [35]. However, due to the influence of the atmospheric blurring effects, the quantitative application of high-spatial resolution data is still restricted to a certain extent. Therefore, atmospheric correction (AC) has to be conducted to promote the quantitative application of remote sensing data, e.g., obtaining surface physical parameters. According to whether it is quantitatively processed, AC can be based on image processing [51] or radiative transfer calculation methods [3], [10], [22], [33], [40], [39], [43], [44], in which the principle of radiative transfer is widely used for AC because of its clear physical meaning. The challenge of AC is usually to obtain high-precision correction parameters, and there are currently three kind of approaches. One is to use the sensor's own information to estimate atmospheric parameters for AC, which is suitable for multispectral medium-resolution images, such as Moderate-Resolution Imaging Spectroradiometer (MODIS) [25], [45].

The second method is to obtain correction parameters from ground-based or other satellite sensor products. In this case, however, the atmospheric conditions often change greatly with time and space, and further cause spatiotemporal matching

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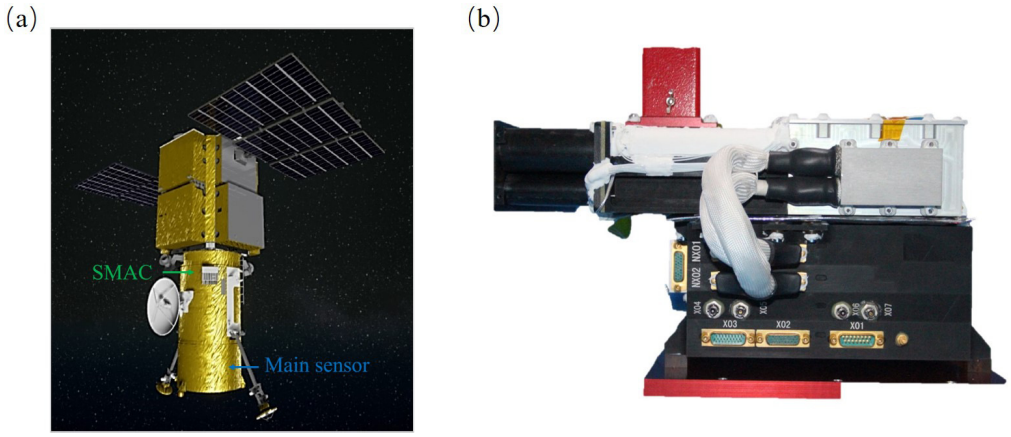


Fig. 1. (a) High-resolution multimode imaging satellite GFDM satellite. (b) Onboard SMAC.

problems, leading to difficulties in meeting the needs of an operational correction. For sensors with high-spatial resolution, as there are usually limited number of channels, it is difficult to use their own measurements to retrieve high-precision atmospheric parameters for AC. Thus, the third method is to obtain the correction parameters by carrying a synchronous atmospheric monitoring instrument on the same satellite platform. This method can not only solve the spatiotemporal matching problem of correction parameters, but also provide higher accurate atmospheric parameters, which representing an important direction of future development for high-spatial resolution satellites [14].

One example is the Earth Obseving-1 (EO-1) satellite, which was launched by the United States (US) of America in November 2000. A linear etalon imaging spectrometer array (LEISA) atmospheric corrector (LAC) was equipped on EO-1 satellite to correct the multispectral imagery for advanced land imager (ALI) [38]. In January 2009, Japan launched the Greenhouse Gases Observing Satellite (GOSAT), which carried a Thermal And Near infrared Sensor for carbon Observation Cloud and Aerosol Imager (TANSO-CAI) sensor for correcting clouds and aerosol effects [24]. Besides, the WorldView-3 satellite launched by US commercial company (DigitalGlobe) in August 2014 also equips with an AC instrument, the Clouds, Aerosols, water Vapor, Ice, and Snow (CAVIS) [1].

In July 2020, the high-resolution multimode imaging satellite [Gao Fen Duo Mo (GFDM)] was launched by the China National Space Administration (CNSA). The GFDM satellite was equipped with one panchromatic and eight multispectral bands with the spatial resolution of 0.42 and 1.6 m, respectively, and it is of great significance in the application of terrestrial optical remote sensing. As the first atmospheric corrector onboard high-spatial resolution satellite with polarization detection, the synchronization monitoring atmospheric corrector (SMAC) was equipped on GFDM, offering time synchronized and field of view overlapped atmospheric measurements to improve the AC of the GFDM main sensor image. The SMAC was developed by the Anhui Institute of Optics and Fine Mechanics, Chinese Academy of Sciences (CAS). The reason why the polarization was integrated is based on that polarized detection is relatively more sensitive to atmosphere, and as a result, the retrieval of aerosol parameters can be benefited from this sensitivity [9], [11], [13], [15], [20], [27], [52].

The performance of the correction parameters provided by the SMAC is the guarantee of the AC of the main sensors. This study introduces the retrieval algorithm of the SMAC atmospheric parameters, including aerosol optical depth (AOD) and column water vapor (CWV), as well as validation of the products with AERosol RObotic NETwork (AERONET) observations. It is arranged as follows. In Section II, we introduce SMAC sensor information; in Section III, we introduce the introduction of SMAC's cloud detection, CWV, and AOD retrieval algorithms; in Section IV, the SMAC measurements under typical atmospheric conditions are illustrated and analyzed, and AC was carried out based on high-spatial resolution multispectrum images of the main sensor; effect of aerosol models and cloud detection is discussed in Section V; Section VI is a conclusion.

II. Satellite and Instrument

A. GFDM Satellite

Fig. 1 shows a diagram of GFDM satellite developed by the China Academy of Space Technology, which was launched into a sun-synchronous orbit on July 3, 2020. GFDM is equipped with a main sensor, an atmospheric corrector SMAC, and other auxiliary equipment. It is the first civilian optical remote sensing satellite in China with spatial resolution reaching submeter level [30]. The main charge coupled device (CCD) sensor has one panchromatic (Pan: 450–900 nm) and eight multispectral bands (B1: 450–520 nm, B2: 520–590 nm, B3: 630–690 nm, B4: 770–890 nm, B5: 400–450 nm, B6: 590–625 nm, B7: 705–745 nm, B8: 860–1040 nm) with the spatial resolution of 0.42 and 1.6 m at nadir, respectively. In addition, GFDM has multiple agile imaging modes, which makes GFDM satellite more efficient in obtaining high-spatial resolution images of targets, which is expected to be widely used and further meets the needs of high-spatial resolution data for the application like disaster mitigation and emergency response.

B. SMAC Sensor

Fig. 1(b) gives an illustration of the SMAC sensor, which utilizes split-aperture detection method to achieve

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TABLE I
CHARACTERISTICS OF THE SMAC SENSOR ONBOARD GFDM SATELLITE

Parameter	Value	Parameter	Value
Spatial resolution	6.7 km	Polarized angle	0°, 60°, 120° (490, 670, 870, 1610 nm) 0°, 60°, 120°, 145° (2250 nm)
Instrument FOV	1.48°	Spectral band	490#, 550, 670#, 870#, 910, 1380, 1610#, 2250# nm
Stokes parameters	I, Q, U	Band width	20, 20, 20, 40, 20, 40, 60,80 nm
Swath width	2 pixels	Rad. Cal. Error	≤ 5% (490, 550, 670, 870, 910 nm) ≤ 6% (1380, 1610, 2250 nm)
Imaging	No	Pol. Cal. Error	≤ 0.01 (DOLP ≥0.02)

Bands with polarization

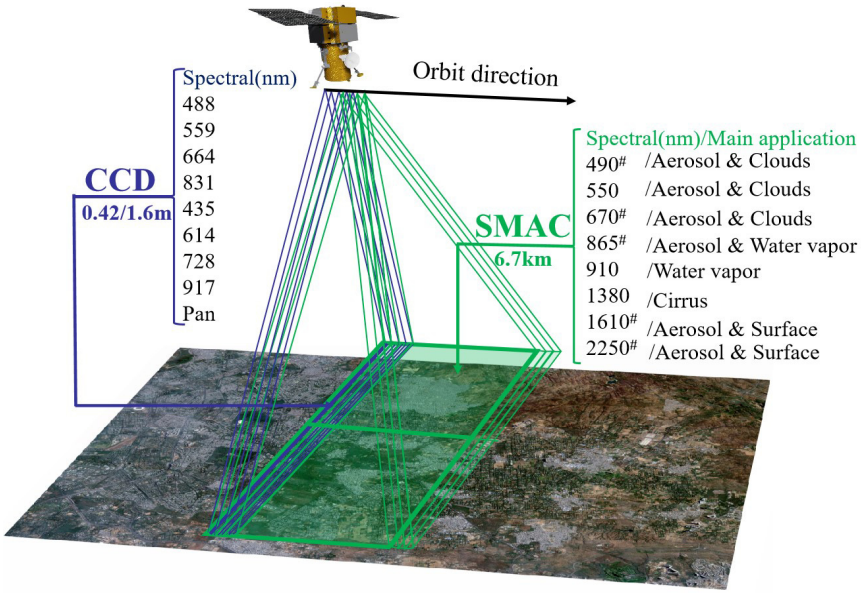


Fig. 2. Diagram of the SMAC and main CCD sensor imaging. The black arrow represents the moving direction of satellite, and the bottom image represents the high-spatial resolution images from main sensor. The spatial resolution of main CCD sensor is 0.42 m for one panchromatic and 1.6 m for eight multispectral bands from visible to near infrared. The green box pixels mean the measurement of SMAC. The spatial resolution of SMAC is 6.7 km with eight wavelength bands from visible to shortwave infrared, and five of them have the capacity of polarization.

simultaneous measurement of each polarimetric channel and eliminate the pseudo-polarization effect caused by time-sharing polarization measurement. The high-precision integrated structure design ensures the accuracy of the field of view (FOV) of each polarimetric detection channel and reduces the polarization measurement error. SMAC has eight wavelength bands from visible to short-wave infrared (490–2250 nm), five channels with polarization detection centered in 490, 670, 870, 1610, and 2250 nm, and the spatial resolution is about 6.7 km. The detailed parameters of SMAC are shown in Table I.

The SMAC uses a dual-pixel shooting mode and synchronously works together with the main CCD sensor. The synchronous observation method of SMAC and the main sensor are shown in Fig. 2, in which the bottom image is a high-spatial resolution image captured by the main sensor, and the green box pixels on the upper layer represent SMAC measurements. In the moving direction of the satellite, SMAC performs multiple (redundant along-track) shooting measurement, while the main sensor is imaging.

III. DATA PRODUCTS AND ALGORITHM

A. Aerosols

The SMAC aerosol inversion algorithm is based on the optimization inversion method, and the main principles are as follows:

1) Forward Model: Based on the optimal estimation (OE) theory [11], the forward model can be defined in the form as

$$\mathbf{y} = \mathbf{F}(\mathbf{x}, \mathbf{b}) + \boldsymbol{\epsilon} \tag{1}$$

where $\mathbf{x}$ represents a state vector to be retrieved, $\mathbf{b}$ represents the aerosol model, $\mathbf{y}$ means an observation vector with corresponding measurements, F means a forward model, and $\boldsymbol{\epsilon}$ is an experimental error term that includes observation noise and forward modeling uncertainty.

For the aerosol model in the forward simulations, a bimodal lognormal function is used to describe the particle size distribution of spherical aerosol particles with the combination of fine-mode and coarse-mode aerosols, and two typical aerosol models including Urban Polluted and Dust are adopted in this

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TABLE II
BASIC PARAMETERS OF AEROSOL MODEL

Aerosol Model	Mode	r_{eff} (μm)	V_{eff}	m_r	m_i
Urban Polluted	Fine-mode	0.156	0.3039	1.43[a]	0.0068[a]
				1.44[b]	0.0053[b]
				1.44[c]	0.0054[c]
	Coarse mode	2.282	0.471	1.56[a]	0.0043[a]
				1.56[b]	0.0022[b]
				1.55[c]	0.0019[c]
Dust	Fine-mode	0.1278	0.427	1.37[a]	0.0044[a]
				1.39[b]	0.0036[b]
				1.41[c]	0.0038[c]
	Coarse mode	1.9387	0.3739	1.53[a]	0.003[a]
				1.55[b]	0.0026[b]
				1.54[c]	0.0025[c]

[a,b,c] The complex refractive index of 490, 670 and 865 nm respectively.

study. The detailed parameters of selected aerosol models are shown in Table II, r_{eff} represents the effective radius, V_{eff} is effective deviation, and m_r and m_i are the real part and imaginary part of complex refractive index at 490, 670, and 865 nm, respectively [53].

The wavelength-dependent AOD can be derived by the columnar volume concentration considering the standard spherical assumption on particle sharps [20], [50], [53]

$$\tau_{\text{a}} = \frac{3V_0^{\text{f}}}{4r_{\text{eff}}^{\text{f}}} Q_{\text{ext}}^{\text{f}} + \frac{3V_0^{\text{c}}}{4r_{\text{eff}}^{\text{c}}} Q_{\text{ext}}^{\text{c}} \tag{2}$$

where V_0^{f} and V_0^{c} mean the columnar volume concentration for the fine- and coarse-mode aerosol, respectively, Q_{ext} means the extinction efficiency factor, and the superscripts f and c represent the fine and coarse modes, respectively. Based on the definition of V_0^{f} and V_0^{c}, we can further have the total aerosol volume $V_0 = V_0^{\text{f}} + V_0^{\text{c}}$ and volume fine-mode fraction $\text{FMF}_V = V_0^{\text{f}}/V_0$.

Unified Linearized Vector Radiative Transfer Model (UNL-VRTM) is used as the forward model in this study [47]. For the sake of simplicity, the top-of-atmosphere (TOA) polarized reflectance $R_{p,\lambda}^{\text{TOA}}$ can be written as [49], [53]

$$R_{p,\lambda}^{\text{TOA}}(\theta_0, \theta_{\text{v}}, \phi) = R_{p,\lambda}^{\text{Atm}}(\theta_{\text{s}}, \theta_{\text{v}}, \phi) + T_{p,\lambda}(\theta_{\text{s}})^{\downarrow} R_{p,\lambda}^{s}(\theta_{\text{s}}, \theta_{\text{v}}, \phi) T_{p,\lambda}(\theta_{\text{v}})^{\uparrow} \tag{3}$$

where θ_{s}, θ_{v}, and ϕ represent the solar zenith angles, viewing zenith angle, and the relative azimuth angle, respectively; $T_{p,\lambda}^{\downarrow}$ and $T_{p,\lambda}^{\uparrow}$ mean the direct transmission term for the directions of upwelling and downwelling, respectively, which can be approximately calculated by following the work of Waquet *et al.* [49]; $R_{p,\lambda}^{s}$ is the surface polarized reflectance. Here, the first term on the right side of (3) means the polarized scattering contribution of molecules and aerosols in the atmosphere, while the second term describes the single reflection of the surface and the transmission directly through the atmosphere.

For the wavelength-independent surface-polarized reflectance, R_p^s can be represented as [28], [29]

$$R_p^s(\theta_0, \theta_v, \phi) = \frac{C\exp(\tan\gamma)\exp(-\text{NDVI})}{4(\cos\theta_0 + \cos\theta_{\text{v}})} \mathbf{F}_{1,2}(\gamma, n_i) \tag{4}$$

where γ means the half of the phase angle, NDVI means the normalized difference vegetation index, $\mathbf{F}_{1,2}(\gamma, n_i)$ represents the polarized component of the Fresnel reflection matrix $\mathbf{F}(\gamma, n_i)$, n_i is the reflective index of the vegetative matter, and C is the only free linear parameter of this model.

2) OE Inversion Framework: For OE inversion, the simplified scalar-valued cost function $J(\mathbf{x})$ can be written as [21], [27]

$$J(\mathbf{x}) = \frac{1}{2}\left[\mathbf{y} - \mathbf{F}(\mathbf{x})\right]^{\text{T}} \mathbf{S}_{\text{y}}^{-1} \left[\mathbf{y} - \mathbf{F}(\mathbf{x})\right] \tag{5}$$

where $\mathbf{S}_{\text{y}}$ means the measurement error covariance matrix, which is usually assumed as a diagonal matrix.

Here, we define the state vector as

$$\mathbf{x} = \left[V_0^{\text{f}}, V_0^{\text{c}}, C\right]^{\text{T}} \tag{6}$$

which only contains two parameters for the aerosols and one parameter for the BPDF. Meanwhile, we have the vector of aerosol model $\mathbf{b}$ as

$$\mathbf{b} = \left[r_{\text{eff}}^{\text{f}}, v_{\text{eff}}^{\text{f}}, r_{\text{eff}}^{\text{c}}, v_{\text{eff}}^{\text{c}}, m_r^{\text{f}}(\lambda), m_i^{\text{f}}(\lambda), m_r^{\text{c}}(\lambda), m_i^{\text{c}}(\lambda)\right]^{\text{T}}. \tag{7}$$

In addition, the measurements vector $\mathbf{y}$ can be represented by

$$\mathbf{y} = \left[R_{p,\lambda_1}^{\text{TOA}}, R_{p,\lambda_3}^{\text{TOA}}, R_{p,\lambda_4}^{\text{TOA}}, R_{p,\lambda_7}^{\text{TOA}}, R_{p,\lambda_8}^{\text{TOA}}\right]^{\text{T}} \tag{8}$$

where λ_1, λ_3, λ_4, λ_7, and λ_8 represent the polarized bands centered in 490, 670, 870, 1610, and 2250 nm, respectively.

In this study, under optimized inversion framework, we used the parameter traversal method based on a lookup table (LUT) instead of the optimized iteration with real-time calculation. And the parameter combination of state vector corresponding to the smallest value function is the optimal parameter combination to be retrieved, and then based on (2), the AOD can be obtained.

B. Water Vapor

The water vapor retrieval is based on channel ratio between the 865 nm window channel and the 910 nm water vapor absorption channel, and the water vapor LUT established by the Second Simulation of the Satellite Signal in the Solar Spectrum (6S) radiation transmission model [46].

Assuming that the surface is a uniform Lambertian and within narrow spectral bands outside of main absorption features of water vapor, the relationship between the apparent reflectance obtained by the satellite and the surface reflectance is as follows [45]:

$$R_{\lambda}^{\text{TOA}}(\theta_{\text{s}}, \theta_{\text{v}}, \phi) = T_g(OG)\left[R_{\lambda}^{\text{Atm}}(\theta_{\text{s}}, \theta_{\text{v}}, \phi) + \frac{T_{\lambda}(\theta_{\text{s}})^{\downarrow} R_{\lambda}^{s}(\theta_{\text{s}}, \theta_{\text{v}}, \phi) T_{\lambda}(\theta_{\text{v}})^{\uparrow}}{1 - S_{\lambda} R_{\lambda}^{s}(\theta_{\text{s}}, \theta_{\text{v}}, \phi)} T_g(\text{H}_2\text{O})\right] \tag{9}$$

where $\rho_{\lambda}^{\text{TOA}}(\theta_{\text{s}}, \theta_{\text{v}}, \phi)$ is apparent reflectance, $R_{\lambda}^{\text{Atm}}(\theta_{\text{s}}, \theta_{\text{v}}, \phi)$ represents atmospheric path reflectance, $Tg(\text{H}_2\text{O})$ and $Tg(OG)$ are the gaseous transmission by water vapor and

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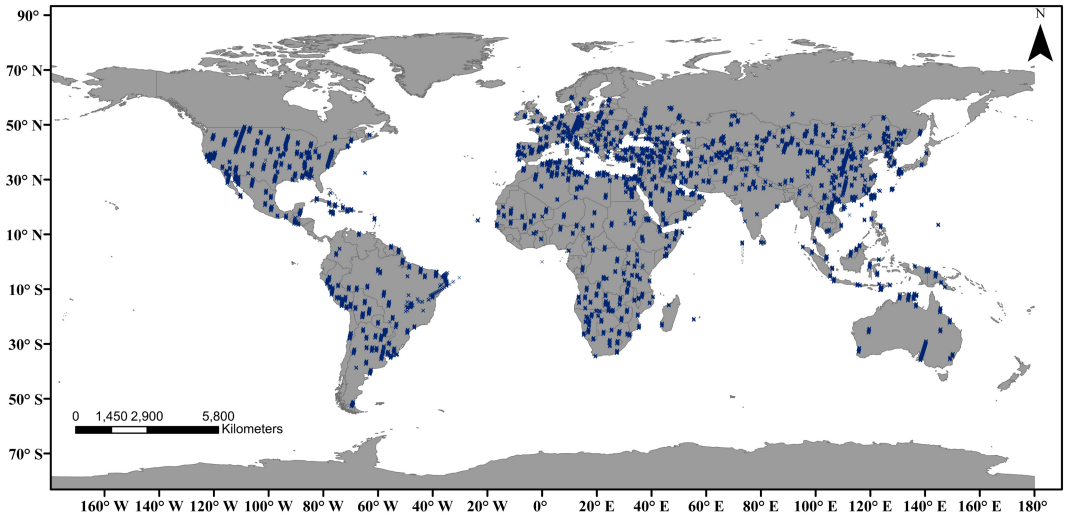


Fig. 3. SMAC track coverage (blue dots) from 15 July to 19 August 2020.

other gases (O_3, CO_2, etc.), $T_\lambda(\theta_s)^\downarrow T_\lambda(\theta_v)^\uparrow$ is the total transmittance, S_λ is the hemispherical albedo, and $\rho_\lambda(\theta_s, \theta_v, \phi)$ is angular surface reflectance.

Furthermore, assuming that the surface reflectance at 865 and 910 nm is similar or changes gently, the ratio of the atmospheric transmittance of these two bands is approximately equal to the ratio of the reflectance of the two bands

$$\frac{R_{910}^{\mathrm{TOA}} - R_{910}^{\mathrm{Atm}}}{R_{865}^{\mathrm{TOA}} - R_{865}^{\mathrm{Atm}}} \simeq \frac{T_{910}(\theta_s)^\downarrow T_{910}(\theta_v)^\uparrow}{T_{865}(\theta_s)^\downarrow T_{865}(\theta_v)^\uparrow}. \tag{10}$$

The ratio of atmospheric transmittance is between 910 and 865 nm on account of the radiative transmission simulation, and the atmospheric column water vapor can be fitted into an empirical formula [2], [5] as follows:

$$U_{\mathrm{cwv}} = \frac{a_0 + a_1 \ln(T) + a_2[\ln(T)]^2}{m} \tag{11}$$

where $T = (T_{910}(\theta_s)^\downarrow T_{910}(\theta_v)^\uparrow)/(T_{865}(\theta_s)^\downarrow T_{865}(\theta_v)^\uparrow)$, $m = 1/(\cos(\theta_s)) + 1/(\cos(\theta_v))$, and $a_0 = 0.1946$, $a_1 = 0.5202$, and $a_2 = 28.11$ are fitting coefficients of empirical formula.

C. Cloud Mask

The existence of clouds has a great influence on the inversion accuracy of key atmospheric parameters; therefore, it is necessary to identify and remove cloud-covered pixels before retrieving AOD and CWV from SMAC level 1 data. In this article, a threshold-based method has been used for cloud masks, and corresponding threshold values are listed empirically from the SMAC data as follows.

1) To detect thick cloud, considering the brightness of thick clouds is generally very large, the TOA reflectance at 490 nm is usually used for cloud judgment [26]

$$R_{490}^{\mathrm{TOA}} > 0.4. \tag{12}$$

2) For cirrus detection, the following threshold judgments are adopted [26]:

$$R_{1380}^{\mathrm{TOA}} > 0.0025. \tag{13}$$

3) For dust case, the Normalized Difference Dust Index (NDDI) is used as [7]

$$\left(R_{2250}^{\mathrm{TOA}} - R_{490}^{\mathrm{TOA}}\right)/\left(R_{2250}^{\mathrm{TOA}} + R_{490}^{\mathrm{TOA}}\right) < 0. \tag{14}$$

4) While for snow detection, the Normalized Difference Snow Index (NDSI) is further considered [17]

$$\left(R_{550}^{\mathrm{TOA}} - R_{1610}^{\mathrm{TOA}}\right)/\left(R_{550}^{\mathrm{TOA}} + R_{1610}^{\mathrm{TOA}}\right) > 0.13. \tag{15}$$

For each SMAC pixel, (12)–(15) are calculated, and it is recognized as cloud contaminated when at least one of the above equation stands.

IV. On-Board Performance Test

A. SMAC Measurements

Fig. 3 illustrates the SMAC orbital distribution from July 15, 2020 to August 19, 2020. It reads from the figure that the measurements of SMAC are distributed globally, excluding high latitude region. In addition to the regular orbit distribution, it can be clearly seen that the large oblique orbit distribution over Brazil region reflects the mobility observation capabilities of the GFDM satellite.

B. First Results and Validation

To test the accuracy of atmospheric parameters retrieved from SMAC measurements under different atmospheric conditions, we present the typical AOD and CWV retrieval results under typical atmospheric conditions in Section IV-B1 and then validate all the retrieval results against AERONET measurement in Section IV-B2. The AERONET is usually considered as "ground truth" for validating satellite-based products [16], [19]. In terms of time matching, at least two AERONET observations are chosen as the average value within 30 min before and after the satellite overpass [37]. In terms of spatial matching, all the valid SMAC retrieval results are averaged within 0.3° of latitude and longitude (about 30 km) centered on the ground site for validation [23].

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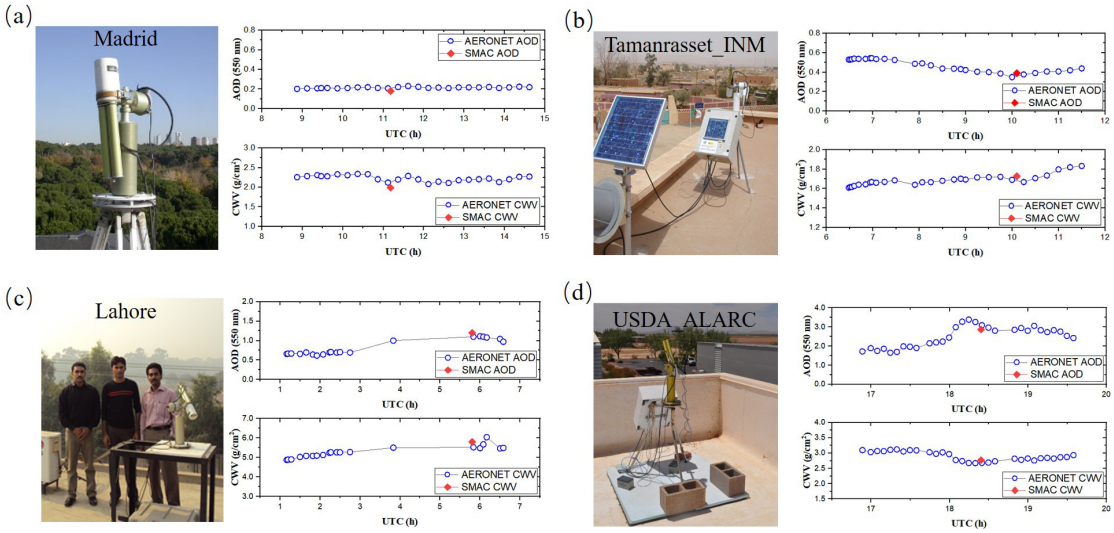


Fig. 4. SMAC AOD and CWV retrieval results at AERONET sites. (a) Madrid site. (b) Tamanrasset_INM site. (c) Lahore site. (d) USDA ALARC site. The blue solid dots mean the time series observations of AERONET, and red diamond represents the products of SMAC.

1) Typical Case Study:

a) Clear atmosphere case: For the clear atmospheric case, the data at AERONET Madrid site (urban surface) located in Spain with the latitude and longitude of 40.452N and 3.724W on July 27, 2020, are used. Validation results are shown in Fig. 4(a) and the AOD at 550 nm of the ground-based observations is 0.21, while the retrieval result of SMAC is 0.18 and the relative error (RE) is 14%. The CWV of the AERONET is 2.21 g/cm^2, while the result of SMAC is 1.99 g/cm^2 and the RE is 10%.

b) Slightly polluted case: As for the slightly pollution case, the data at the AERONET Tamanrasset_INM site located in Algeria with the latitude and longitude of 22.79 N and 5.53 E on August 2, 2020, are employed. The surface type is desert. As shown in Fig. 4(b), the AOD of the AERONET is 0.4, while the retrieval result of SMAC is 0.39 and the RE is 3%. The CWV of the AERONET site is 1.72 g/cm^2, while the retrieval result of SMAC is 1.7 g/cm^2 and the RE is only 1%.

c) Moderate pollution case: Regarding the moderate pollution case, the data on July 26, 2020, from the AERONET Lahore site located in the Pakistan are employed. The latitude and longitude of the site are 31.48N and 74.264W, respectively, and the surface type is urban. As shown in Fig. 4(c), the AOD of AERONET site is 1.08, while the retrieval result of SMAC is 1.2 and the RE is 11%. The CWV of AERONET site is 5.6 g/cm^2, while the retrieval result of SMAC is 5.8 g/cm^2 and the RE is 4%.

d) Heavy pollution case: Regarding the heavy pollution case, the data on August 2, 2020, from the AERONET USDA_ALARC site located in the western United States are employed, which is affected by the extraordinary fire season of 2020 [18]. The latitude and longitude of the site are 33.077N and 111.973W, respectively, and the surface type is bare land. Due to the wildfire, as shown in Fig. 4(d), the AOD of AERONET site is 2.9, while the retrieval result of SMAC is 2.87 and the RE is 1%. The CWV of AERONET site is 2.7 g/cm^2, while the retrieval result of SMAC is 2.8 g/cm^2 and the RE is 4%.

2) Validation Against AERONET: Due to the reduced major axis (RMA) can simultaneously account for errors in both dependent and independent variables [4], [8], which was used to calculate the slope and intercept between SMAC and AERONET products. Fig. 5 illustrates the validation of SMAC AOD and CWV results from July 15, 2020 to August 19, 2020 over global AERONET sites. There are a total of 31 matchup points (N) for all AOD and CWV [Fig. 5(a) and (c)], and 29 AOD matchup points of them are less than 1 [Fig. 5(b)]; the correlation coefficient (R) values are 0.96 and 0.98 for all AOD and CWV, respectively; the slope and intercept of fit line are 0.99, 0.04, 1.08, and −0.4, respectively; the corresponding root-mean-square error (RMSE) values are 0.13 and 0.26, respectively; even the difficulties on validation point match up as statement in Section IV-B; these preliminary batch of validation results suggests that both AOD and CWV are in good agreement versus ground-based observations. However, for the case of AOD less than 1, it is observed that quite a few AOD points are overestimated than the 1:1 line, and the reason was thought to be that SMAC is not imaging, while the spatial resolution is 6.7 km and the small broken clouds (e.g., thin cloud contamination) may not be totally identified with the current cloud masking algorithm. This suggests that the retrieved AOD results may bring more errors in the AC method during low aerosol loadings than high case; more discussion can be found in Section V-B.

C. Atmospheric Correction

Absorption and scattering from atmospheric parameters can blur and reduce the contrast of surface features of satellite images, and it is essential to carry out AC before performing visual interpretation and quantitative research. For high-spatial resolution images, AC mainly removes the atmospheric intrinsic reflectance and adjacency effect,

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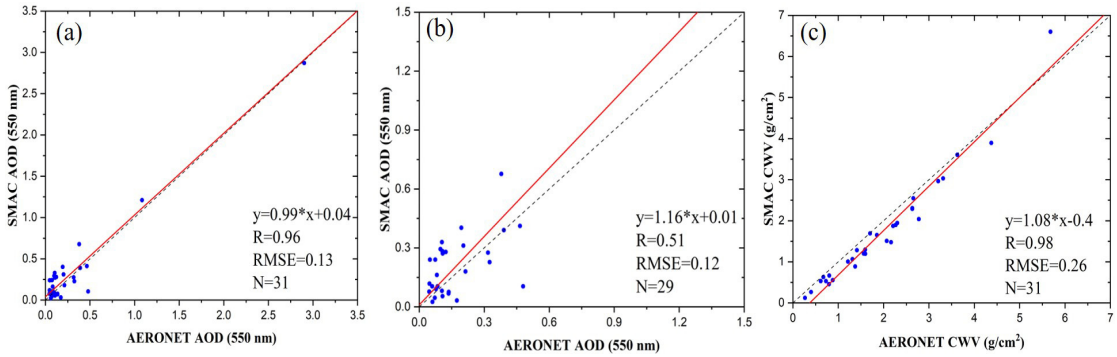


Fig. 5. Validation of SMAC all (a) AOD, (b) AOD less than 1, and (c) CWV versus AERONET observations from July 15 to August 19, 2020. The corresponding wavelength of the AOD value is 550 nm. The dashed line and solid line are the 1:1 line and fit line, respectively. The statistics of the linear fitting result, including slope, intercept, R, RMSE, and number of matchup points (N), are listed in the bottom right corner of each scatter plot.

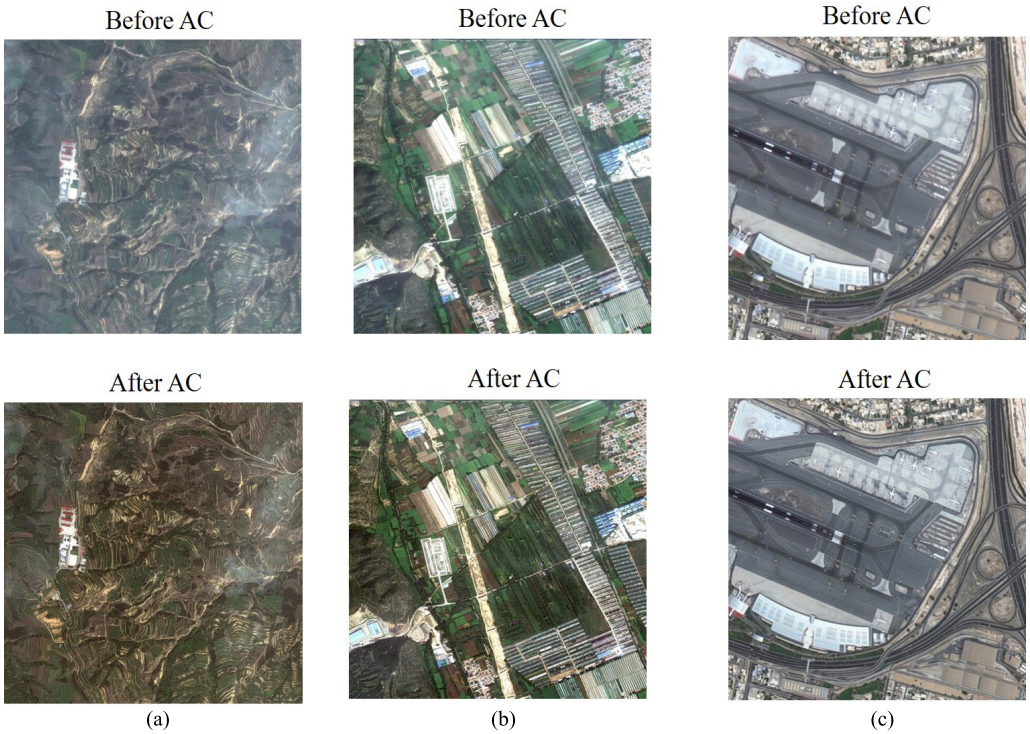


Fig. 6. True color images of main CCD sensor of GFDM satellite before and after atmospheric correction (AC). Images (a) and (b) are located in Qinghai, China, and image (c) is located in Dubai, United Arab Emirates. The AC parameters are retrieved from SMAC sensor with AOD of 0.4, 0.25, and 0.09, and the CWV of 1.2, 1.33, and 4.21 g/cm^2 for images (a)–(c), respectively.

which is the contribution of neighboring pixels to the radiance of the lines-of-sight pixel [41], [48]. In this study, AC method is as follows: First, LUT is computed by 6SV and sufficiently representative of all reasonably likely atmospheric scenarios and satellite observations, which only includes the atmospheric parameters R_λ^{Atm}, $T_\lambda(\theta_s)^{\downarrow}T_\lambda(\theta_v)^{\uparrow}$, and S_λ for different θ_s, θ_v, ϕ, CWV, and AOD; second, assuming that the surface is of uniform Lambertian, and the atmospheric parameters in LUT are interpolated according to the observation geometry parameters, CWV and AOD, then the initial surface reflectance is retrieved, which contain the contribution of adjacency effect, can be canulated by (9) [43]; third, adjacency effect correction is carried out based on the method of radiative transfer and gets the true surface reflectance [36], [43], [41]. The detailed information of AC is shown in the appendix.

In order to illustrate the performance of the synchronous AC parameters provided by the SAMC, three cases of the main CCD sensor are analyzed in this part. We use true color images (composed of B3–B1) considering that they are more severely affected by the atmospheric effects. As shown in Fig. 6(a)–(c), true color images before AC are observed from main CCD sensor on the upper panel, and it can be seen that the features of surface (buildings, roads, vegetations, etc.) are blurring caused by the atmospheric effects. Then, we carry out AC for main CCD sensor images based on the radiative transfer method, using the synchronous AOD and CWV

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TABLE III

QUALITY INDICATORS [CONTRAST (σ_c) AND THE LEGIBILITY (σ_l)] OF MAIN CCD SENSOR IMAGES OF FIG. 6(a)–(c) BEFORE AND AFTER AC

		Image (a)			Image (b)			Image (c)		
Band	Evaluating indicator	Before AC	After AC	Improvement (%)	Before AC	After AC	Improvement (%)	Before AC	After AC	Improvement (%)
B3	σ_c	0.23	0.33	10	0.45	0.52	7.0	0.35	0.37	2.0
	σ_l	0.06	0.10	4.0	0.11	0.16	4.0	0.09	0.11	2.0
B2	σ_c	0.20	0.34	14	0.37	0.48	10	0.34	0.36	1.0
	σ_l	0.06	0.10	4.0	0.11	0.16	5.0	0.09	0.11	2.0
B1	σ_c	0.18	0.41	23	0.39	0.57	18	0.33	0.35	2.0
	σ_l	0.06	0.09	4.0	0.11	0.16	5.0	0.09	0.11	2.0

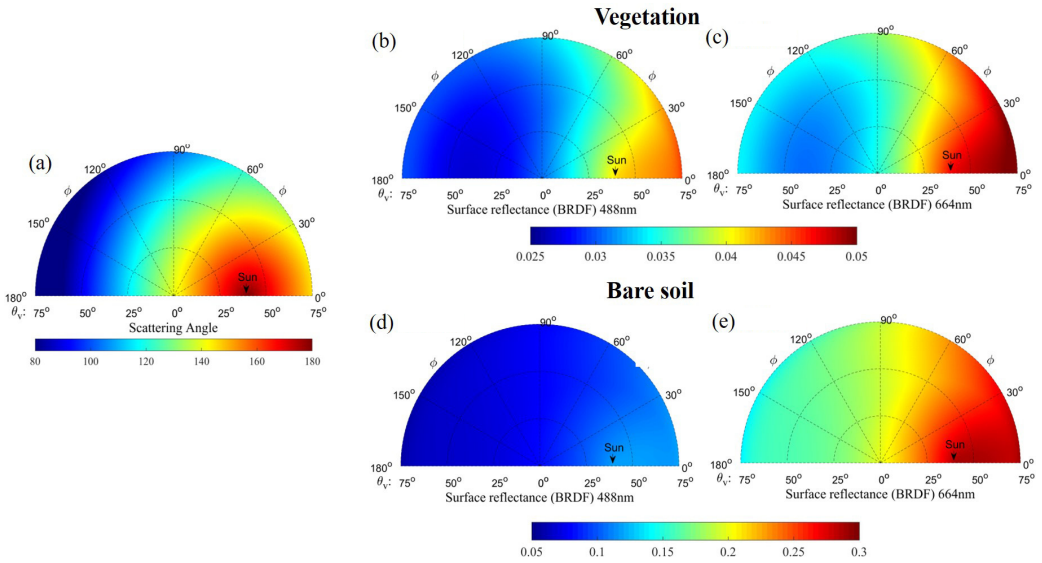


Fig. 7. (a) Polar plot of scattering angle Θ, the radius stands for a change in θ_v from 0° to 75°, and the polar angle ϕ indicates a change from 0° to 180°. When $\phi = 0°$, this geometry represents that the observer and the sun face the same direction and are located on the same side of the main plane. This convention is used for all polar plots in this study. (b)–(e) Polar plots of the angular distribution of the BRDF. (b) 448 nm and (c) 664 nm on vegetation using the same color bar. (d) 448 nm and (e) 664 nm on bare soil using the same color bar.

parameters retrieved by SMAC; the AOD at 550 nm is 0.4, 0.25, and 0.09; and the CWV is 1.2, 1.33, and 4.21 g/cm^2 for cases (a)–(c), respectively, and the true color images after AC are shown in the bottom panel of Fig. 6. Next, the image quality indices [e.g., the Contrast (σ_c) and the Legibility (σ_l)] are calculated from the following equations for images before and after AC (16) and (17), as shown at the bottom of the next page: where m and n are the width and height of images, respectively. $f(x, y)$ is the pixel value in the row x and column y. The results are shown in Table III. It can be seen clearly that all values of σ_c and σ_l of B3–B1 have been improved after AC, while σ_c of Fig. 6(a) is mostly improved (increment of 10%, 14%, and 23%, respectively). σ_l of Fig. 6(b) is mostly improved (increment of 4%, 5%, and 5%, respectively).

V. DISCUSSION

A. Effect of Aerosol Models

In Section III-A1, two aerosol models (Urban Polluted and Dust) are employed to retrieve aerosol microphysical parameters (V_0^f and V_0^c), and then the AOD is calculated based on (2). For other aerosol parameters (such as the complex refractive index, effective radius, and effective deviation), they are all predefined as shown in Table II. However, aerosol is highly variable both in temporal and spatial scales in reality [12]. The effect of these predefined aerosol model parameters might be an important issue to develop the operational algorithm of SMAC in the next works.

To analyze the influence of the aerosol model on AC, we perform the following simplified numerical test. First, two visible bands centered in 488 and 664 nm over two typical surface types (vegetation and bare soil) were used to simulate the TOA reflectance, with different air quality levels (AOD true = 0.25 and 0.8 at 550 nm) and observation geometries ($\theta_s = 40°$, $\theta_v = 0°$ to 75°, and $\phi = 0°$ to 180°). Fig. 7(a) shows the polar plot of scattering angle Θ, the radius stands for a change in θ_v from 0° to 75°, and the polar angle ϕ indicates a change from 0° to 180°. When $\phi = 0°$, this geometry represents that the observer and the sun face the same direction and are located on the same side of the main plane [20]. Correspondingly, the polar plots of the surface reflectance (SR) angular distribution are shown in Fig. 7(b)–(e), in which the surface parameters of bidirectional reflectance distribution function (BRDF) are adopted from Li *et al.* [27], Then,

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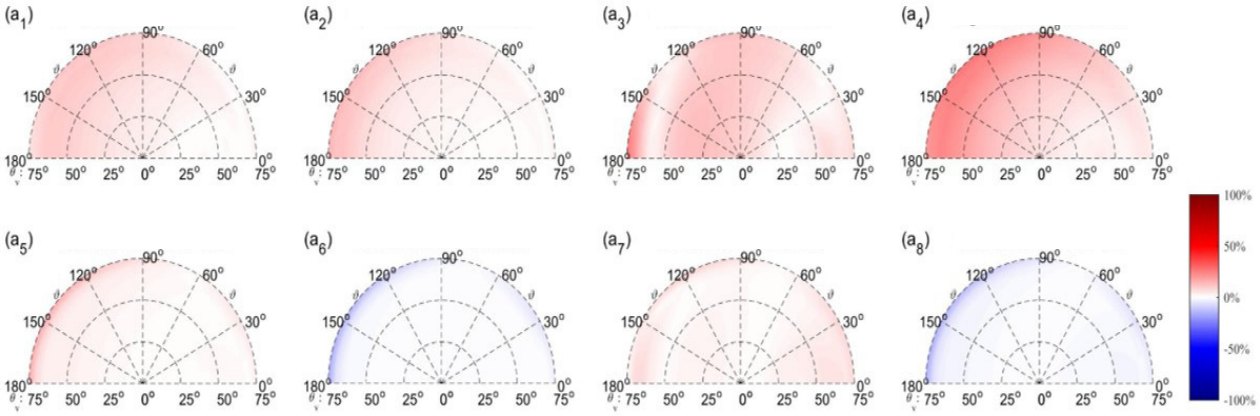


Fig. 8. Polar plot of the relative error between $R_\lambda^{\text{true_}s}$ and $R_\lambda^{\text{bias_}s}$, ($a_1$)–($a_4$) is over the vegetation and (a_5)–(a_8) is over the bare soil, and the band and AOD are shown in the subtitle. Red in the color bar means $R_\lambda^{\text{bias_}s}$ is higher than $R_\lambda^{\text{true_}s}$, and blue means $R_\lambda^{\text{bias_}s}$ is lower than $R_\lambda^{\text{true_}s}$.

TOA reflectance at 488 and 664 nm was simulated using the Urban Polluted aerosol model with $\text{FMF}_V = 0.8$, and the other aerosol parameters are from Table II. After that, we try to perform the AC based on the TOA reflectance using a "wrong" aerosol model (i.e., Dust model with $\text{FMF}_V = 0.2$) and obtained the "biased" surface reflectance ($R_\lambda^{\text{bias_}s}$) under different surfaces types and air quality levels. Next, as shown in Fig. 8, the relative error (RE, %) of true surface reflectance ($R_\lambda^{\text{true_}s}$) and $R_\lambda^{\text{bias_}s}$ were calculated, red in the color bar means $R_\lambda^{\text{bias_}s}$ is higher than $R_\lambda^{\text{true_}s}$, and blue means $R_\lambda^{\text{bias_}s}$ is lower than $R_\lambda^{\text{true_}s}$. It can be seen that the "wrong" aerosol model used for the AC has different effects on obtaining the surface reflectance over different surface and air quality levels. In general, the AC accuracy of surface reflectance over vegetation is more affected by aerosol model error than that of bare soil, because the proportion of aerosol in TOA reflectance is larger than that of surface reflectance. For example, if the band is 664 nm and AOD = 0.8 (a_4) over vegetation, the RE can be reached 26% in the small Θ. However, the model error has little effect on the bare soil (a_5–a_8), and most of the RE are less than 5%.

The numerical experimental results indicate that the aerosol model is important for AC. However, according to previous studies [20], [27] on the information content analysis of the total degrees of freedom for signal in the aerosol retrieval, it may be hard to retrieve all aerosol model parameters simultaneously based on current SMAC configuration. Thus, adding more and realistic predefined aerosol models in the inversion framework should be considered in the next work, together with the potential improvement of hardware configuration (e.g., bands and polarization) in the future.

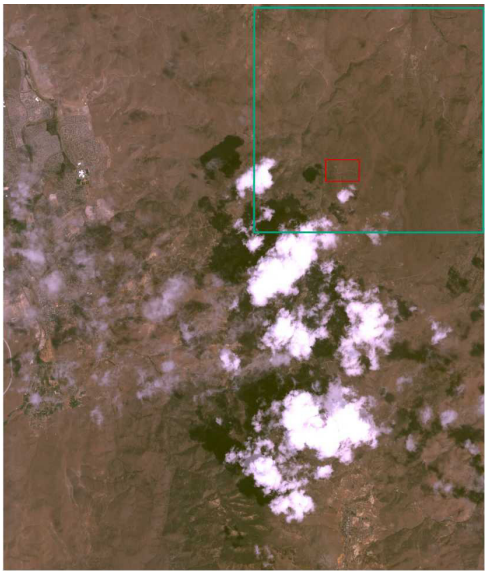


Fig. 9. True color image of main CCD sensor of GFDM on July 22, 2020, which is located on western United States, and the latitude and longitude of center image are 119.668°W and 39.377°N, respectively. Write objects are clouds. The green square represents one pixel of the SMAC (6.7 km × 6.7 km), and it can be seen that some fragmented clouds and thin clouds were coupled into the SMAC pixel. The red square represents a region of interest as shown in Fig. 10.

B. Cloud Detection

The accuracy of cloud detection affects the accuracy of aerosol parameter retrieval, because the miss identification of any small broken and thin clouds will alter the satellite observed TOA reflectance versus the pure cloud-free

$$\sigma_c(x,y) = \frac{\sqrt{\frac{1}{m*n}\sum_{y=1}^{m}\sum_{x=1}^{n} f^2(x,y) - \left(\frac{1}{m*n}\sum_{y=1}^{m}\sum_{x=1}^{n} f(x,y)\right)^2}}{\frac{1}{m*n}\sum_{y=1}^{m-1}\sum_{x=1}^{n-1} f(x,y)} \tag{16}$$

$$\sigma_l(x,y) = \frac{\sum_{y=1}^{m-1}\sum_{x=1}^{n-1}(|f(x,y) - f(x,y+1)| + |f(x,y) - f(x+1,y)|)}{m*n} \tag{17}$$

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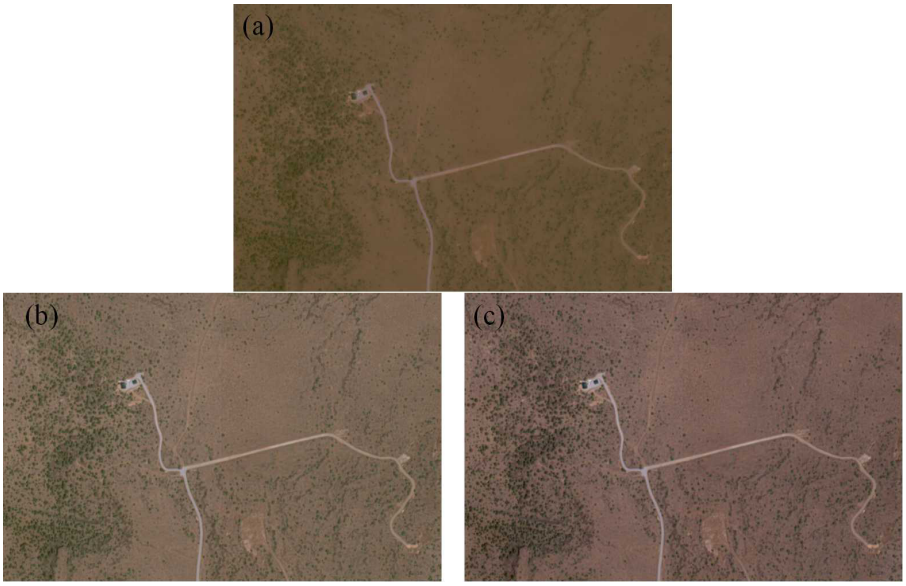


Fig. 10. (a) True color image before AC, which comes from the red square in Fig. 9. (b) and (c) True color image after AC using the AOD and CWV products from the AERONET and SMAC cloud contamination pixels.

assumption in the aerosol retrieval. For the 6.7-km spatial resolution of SMAC, fragmented clouds and thin clouds could be coupled into SMAC pixels in some complex situations using the current threshold judgment method. For example, as shown in Fig. 9, the green square represents an SMAC pixel, and some fragmented clouds and thin clouds were coupled into this SMAC pixel. In this case, the mean AOD and CWV products retrieved from this SMAC pixel are 0.447 at 550 nm and 1.176 g/cm^2, respectively. However, the AOD and CWV products from the AERONET Univ_of_Nevada-Reno (119.814°W, 39.541°N) site are 0.107 and 1.562 g/cm^2, respectively. Thus, if the pixels contaminated by clouds are used for atmospheric parameters retrieval, it could cause large retrieval errors, and then affect the accuracy of AC. As shown in Fig. 10, (a) is the true color image before AC, which comes from the red square in Fig. 9, and (b) and (c) are the true color images after AC using the AOD and CWV products from the AERONET and SMAC cloud contamination pixels, respectively. It can be seen that the surface features, such as the green trees and brown soil, of image (b) are more realistic than those of image (c). Surface features of image (c) seem more darker due to the over-AC using the overestimated AOD. Thus, taking into account that the main CCD sensor has the characteristics of high-spatial resolution, it is possible to combine the spatial characteristics of the main CCD sensor and the threshold judgment method of SMAC to improve the cloud detection performance in the future studies.

VI. Conclusion

This article introduces the SAMC on-board GFDM satellite. We demonstrate the SAMC data during the on-orbit test and introduce SAMC atmospheric parameters retrieval algorithms, i.e., a LUT method with the optimized inversion framework for AOD, and a dual-channel ratio method for CWV. We selected four typical cases as a test, and the RE for AOD is 14%, 3%, 11%, and 1%, respectively, and for CWV, the RE is 10%, 1%, 4%, and 4%, respectively. In addition, the preliminary AOD and CWV results were validated against with global AERONET sites. The validation results show that the retrieval results of SMAC are in good agreement with those from the ground-based observations. Furthermore, the AC was carried out for three typical cases of main CCD sensor's high-spatial resolution images based on the radiative transmission method by using the atmospheric parameters from SMAC. The image quality index (contrast and legibility) of the high-spatial images at three bands (B3–B1) is greatly improved after AC. As the conclusion, our initial results show that the SAMC has potentials to provide synchronized and accurate AOD and CWV products, which can be applied to the AC of the main sensor. The influence on the effects of aerosol model and the cloud detection is also discussed for future improvement of the synchronized atmospheric corrector.

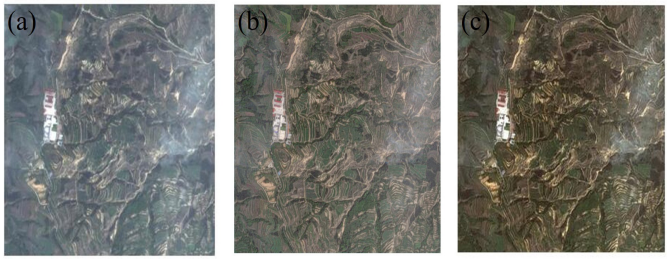


Fig. 11. True color images of (a) the uncorrected satellite measurement (R_λ^{TOA}), (b) the atmospheric blurring effect corrected surface reflectance (R_λ^{s}), and (c) the atmospheric blurring effect plus adjacency effect corrected surface reflectance ($R_\lambda^{\mathrm{true_}s}$) corresponding to Fig. 6(a).

Appendix
Purely Physical AC Method

Assuming a uniform Lambertian surface under plane-parallel atmosphere, the upward spectral reflectance observed by satellite at TOA is usually presented as a function of successive orders of radiation interactions in the surface–atmosphere

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$$R_\lambda^{\mathrm{TOA}} = T_g(OG)\left[R_\lambda^{\mathrm{Atm}} + \frac{T_\lambda(\mu_s)^{\downarrow} R_\lambda^{\mathrm{true_}s} e^{-\tau/\mu_v} + \langle R_\lambda^{\mathrm{true_}s}\rangle T_\lambda(\mu_s)^{\downarrow} t_d(\mu_v)}{1 - S_\lambda \langle R_\lambda^{\mathrm{true_}s}\rangle} T_g(\mathrm{H_2O})\right] \quad \text{(A.2)}$$

coupled system [3], [43]

$$R_\lambda^{\mathrm{TOA}} = T_g(OG)\left[R_\lambda^{\mathrm{Atm}} + \frac{T_\lambda(\mu_s)^{\downarrow} R_\lambda^{s} T_\lambda(\mu_v)^{\uparrow}}{1 - S_\lambda R_\lambda^{s}} T_g(\mathrm{H_2O})\right] \quad \text{(A.1)}$$

where μ_s and μ_v are the cosine of the solar zenith angle and viewing zenith angle, respectively, R_λ^{TOA} is apparent reflectance (satellite measurements), $T_g(\mathrm{H_2O})$ and $T_g(OG)$ are the gaseous transmission by water vapor and other gases (O_3, CO_2, etc.), R_λ^{Atm} represents atmospheric path reflectance, $T_\lambda(\mu_s)^{\downarrow} T_\lambda(\mu_v)^{\uparrow}$ is the bidirectional total transmittances, S_λ is the hemispherical albedo, and R_λ^{s} is surface reflectance. It should be noted here that R_λ^{s} is still not the true surface reflectance of the target pixel in the case of nonuniform surface (which is the most cases except for uniform desert, grassland, etc.).

When considering the nonuniform surface, the pixels' adjacency effect (i.e., the contamination of pixel values due to environmental contribution coming from surrounding pixels through atmospheric refraction) should be considered. The signal at the TOA can be rewritten by decoupling the surface reflectance coming directly from the target ($R_\lambda^{\mathrm{true_}s}$) and those coming from adjacent areas ($\langle R_\lambda^{\mathrm{true_}s}\rangle$) and then scattered to the sensor [43] (A2), as shown at the top of the page where τ is the atmospheric optical depth, $e^{-\tau/\mu_v}$ is the direct upward transmission, $t_d(\mu_v)$ is the diffuse upward transmission, and $t_d(\mu_v) = T_\lambda(\mu_v)^{\uparrow} - e^{-\tau/\mu_v}$. Due to the small value of $S_\lambda\langle R_\lambda^{\mathrm{ture_}s}\rangle$ and $S_\lambda\langle R_\lambda^{s}\rangle$, it could be considered that $1/(1 - S_\lambda\langle R_\lambda^{\mathrm{true_}s}\rangle) \cong 1/(1 - S_\lambda\langle R_\lambda^{s}\rangle)$, and then $R_\lambda^{\mathrm{true_}s}$ and R_λ^{s} are related to the following equation [43]:

$$R_\lambda^{s} = R_\lambda^{\mathrm{true_}s}\frac{e^{-\tau/\mu_v}}{T_\lambda(\mu_v)} + \left\langle R_\lambda^{\mathrm{true_}s}\right\rangle \frac{t_d(\mu_v)}{T_\lambda(\mu_v)}. \quad \text{(A.3)}$$

Here, $\langle R_\lambda^{\mathrm{true_}s}\rangle$ is computed from a subzone of $(2n+1)\times(2n+1)$ pixels of the original image centered on the pixel to be corrected and the formula is as follows:

$$\left\langle R_\lambda^{\mathrm{true_}s}\right\rangle = \sum_{j=-n}^{n}\sum_{i=-n}^{n} f(r(i,j))R_\lambda^{s}(i,j) \quad \text{(A.4)}$$

where $r(i,j)$ representing the distance between the pixels (i,j) and the center of the zone and $f(r(i,j))$ is the atmospheric point spread function [43], [40]. R_λ^{s} is used instead of $R_\lambda^{\mathrm{true_}s}$ since the latter is not available, and the error introduced can be reduced by using successive iterations but is small [36]. Correspondingly, the surface reflectance R_λ^{s} can be obtained from

$$R_\lambda^{\mathrm{true_}s} = \frac{R_\lambda^{s} T_\lambda(\mu_v) - \left\langle R_\lambda^{\mathrm{true_}s}\right\rangle t_d(\mu_v)}{e^{-\tau/\mu_v}}. \quad \text{(A.5)}$$

As an example, Fig. 11 shows the true color images of R_λ^{TOA} (i.e., uncorrected satellite measurement), R_λ^{s} (i.e., the atmospheric blurring effect corrected), and $R_\lambda^{\mathrm{true_}s}$ (i.e., the atmospheric blurring effect plus adjacency effect corrected), which corresponds to the results of Fig. 6(a).

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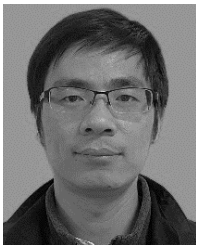
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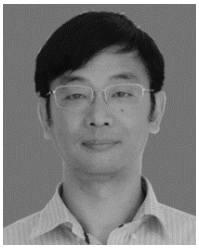
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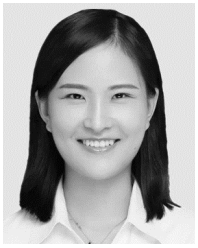
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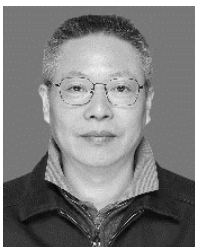
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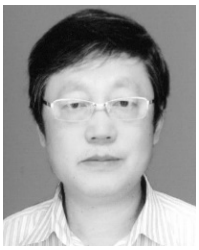
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